

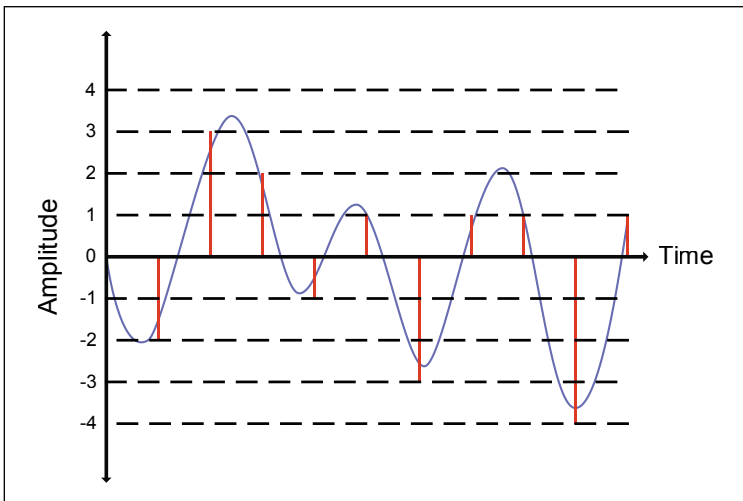
abstraction. Of course, there's a third type of fundamental Digit, but that's us, and we don't like to brag.

Analog signals are continuous, their curve is smooth, whereas digital signals have steps, which represent discrete values. Of course, sound, like most other real world signals, always comes into existence as an analog signal. Then how does the audio become digital? The leap from analog to digital is magic that happens inside an ADC (analog to digital converter) and similarly, a DAC (digital to analog converter) performs the inverse function. It processes the digital audio into a signal that a speaker can play, perhaps after some amplification. ADCs and DACs are vital components of electronic technology and signal processing. Read on to understand how this all works!

Analog to Digital (A/D) conversion

An ADC is a device that takes as an input an analog signal and outputs binary values. It is usually implemented directly in an integrated circuit (IC) like the MCP3551 or ADC0808IC. There are two main aspects of the conversion process that are at odds with each other: Speed and Accuracy. For reference, take a look at the Illustrative Graph, which has been exaggerated for easily conveying an understanding of the process.

In blue we have the analog signal which is being passed into the ADC. This signal is continuous in time and amplitude. To convert this into a digital



The Illustrative Graph - for illustrative purposes only!